Diseases of Beet root

1.Seedling diseases:

A. Damping-off - *Pythium aphanidermatum*, *P. ultimum*

B. Root rot - ***Rhizoctonia solani*** (PS. *Thanatephorus cucumeris*)

C. Black root rot - *Aphanomyces cochlioides*

D. Seed rot and blight - ***Phoma betae***

2. Cercospora leaf spot: *Cercospora beticola*

3. Rhizoctonia root rot and blight – *Rhizoctonia solani*

(PS. *Thanatephorus cucumeris*)

4. Phoma blight and heart rot - *Neocamarosporium betae*

(syns. *Phoma betae*, *Pleospora betae*)

5. Sclerotium root rot - *Sclerotium rolfsii*

6. Beet yellows - *Beet yellows virus* (BYV), *Beet Mild Yellowing Mosaic Virus* (BMYV)

Seedling diseases

1. Heavily infested fields generally appear in patches of different sizes and often in **low lying spots**.
2. However, when conditions are very favourable for disease incidence, the plants throughout the field may exhibit disease symptoms
3. **General symptoms are poor emergence, very uneven growth, dead seedlings and reddish discoloration of above-ground plant parts**

Seedling diseases

Damping off

Wire-stem symptom

4. Water-soaked and necrotic area just below or at the soil line

5,6 . Wire-stem symptom

6,7. Infected roots may rot off just below the soil surface and result in plant death and a reduced stand.

- **Seed decay and damping-off disease**
- **Seed decay** - Seed balls of table beets become infected and decayed prior to germination.
- Pre emergence damping off - Very young seedlings may also become infected and die before they can emerge above the soil surface
- Post emergence damping off - Emerged healthy seedlings may become infected and exhibit a water-soaked and necrotic area just below or at the soil line .
- The latter type of infection may result in wilting, collapse, and death of severely infected seedlings

Wire-stem symptoms / post emergence damping-off stage

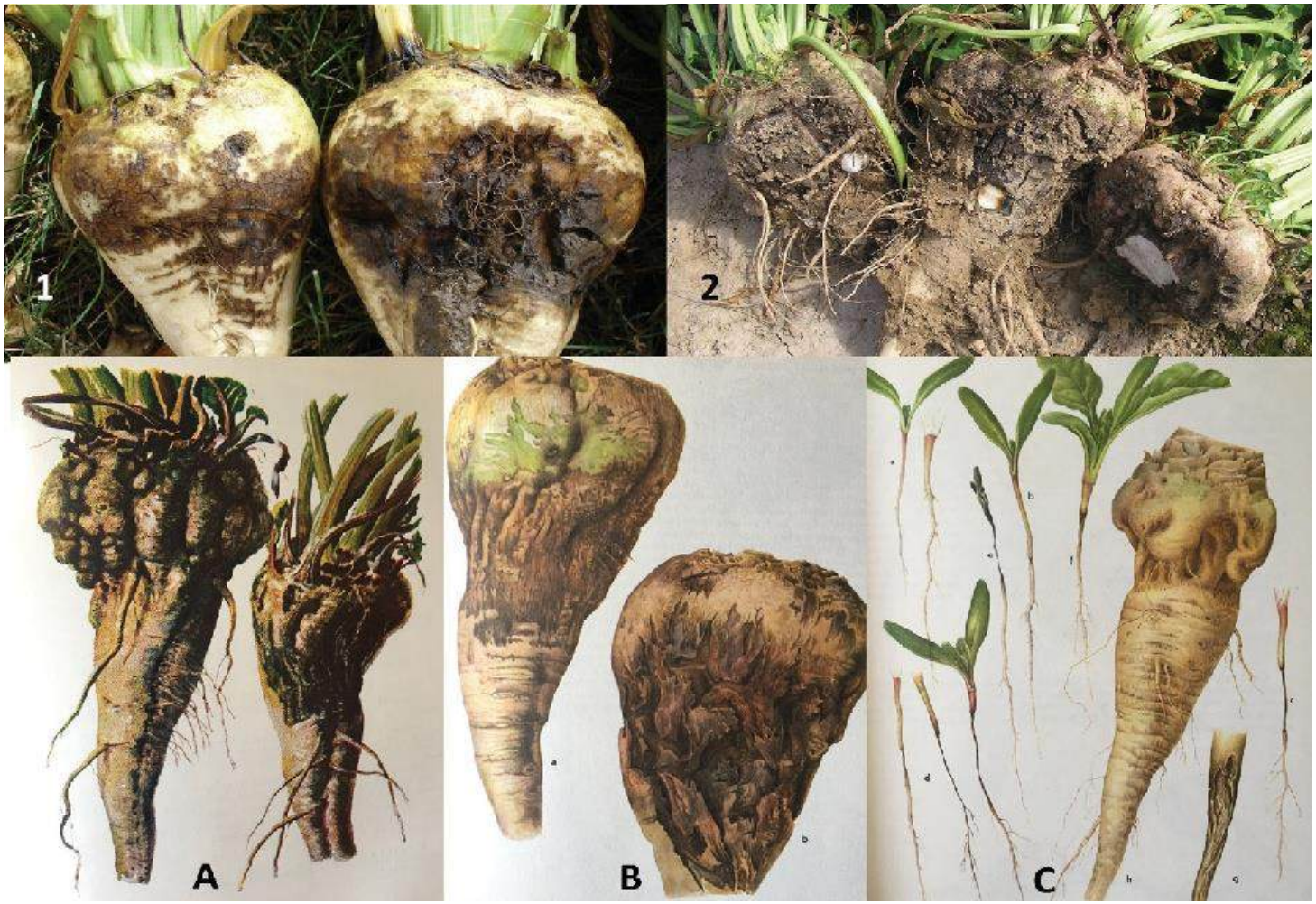
- In 2- to 4-week-old infected seedlings that survive, however the **stem and main root regions become brown to black partially or completely shrivelled; giving them a thread-like appearance** (wire-stem symptom).

- Seedlings with wire-stem symptoms **may have normal branching fibrous root systems**, or roots that are brown and at different stages of rotting.

- Severely infected plants are **stunted and reddish-purple.**

- lf plants are stressed and the infection progresses, **infected roots may rot off** just below the soil surface, and result in plant death and a reduced stand.

Black root rot - *Aphanomyces cochlioides*



Symptoms of Aphanomyces root rot (1, 2, A,B) , (C) seedling damping-off

Black root rot - *Aphanomyces cochlioides*

Symptoms

Root rot two phases – Chronic and acute phase
Acute phase is referred to as **black root,** induces seedling damping off several weeks after emergence

- Post emergence damping off : **Greyish water soaked spots** on stem near the soil line, later turn to gray to black , stem become thin thread like, cotyledons seldom die unlike the symptoms caused by Pythium, Rhizoctonia
- **Chronic root rot phase is more common and occurs on the plant early in the season**
- Affected plants show **temporary wilt symptoms** (unlike Rhizoctonia root rot) - leaves may become **yellow, scorched appearance, brittle**

Black root rot - *Aphanomyces cochlioides*

Symptoms contd…

- Root symptoms any where on **tap root but often occurs towards the distal end**
- As **yellowish brown water soaked lesions** , **later penetrate to the interior tissues which become dry and necrotic**
- If conditions favourable for plant growth, plants may recover and grow, how ever many **roots may still exhibit varying degrees of root distortion and scarring**
- In case of **severe disease incidence tap root completely destroyed** leaving little except the crown

Rhizoctonia root rot & blight/crown rot of beet root – *R. solani*

Rhizoctonia root rot & blight/crown rot of beet root – *R. solani*

Fig. 20. Permanent wilting, collapse of leaves and petioles, and death of plants infected with *R. solani*.

Fig. 17. Root rot phase of Rhizoctonia root rot showing tip rot symptoms and disease progression from the distal tip toward the crown.

Fig. 18. Circular to oval lesions coalescing on tap root, characteristic of Rhizoctonia root rot. Note lack of evidence of infection in crowns.

Rhizoctonia root rot & blight/crown rot of beet root – *R. solani*

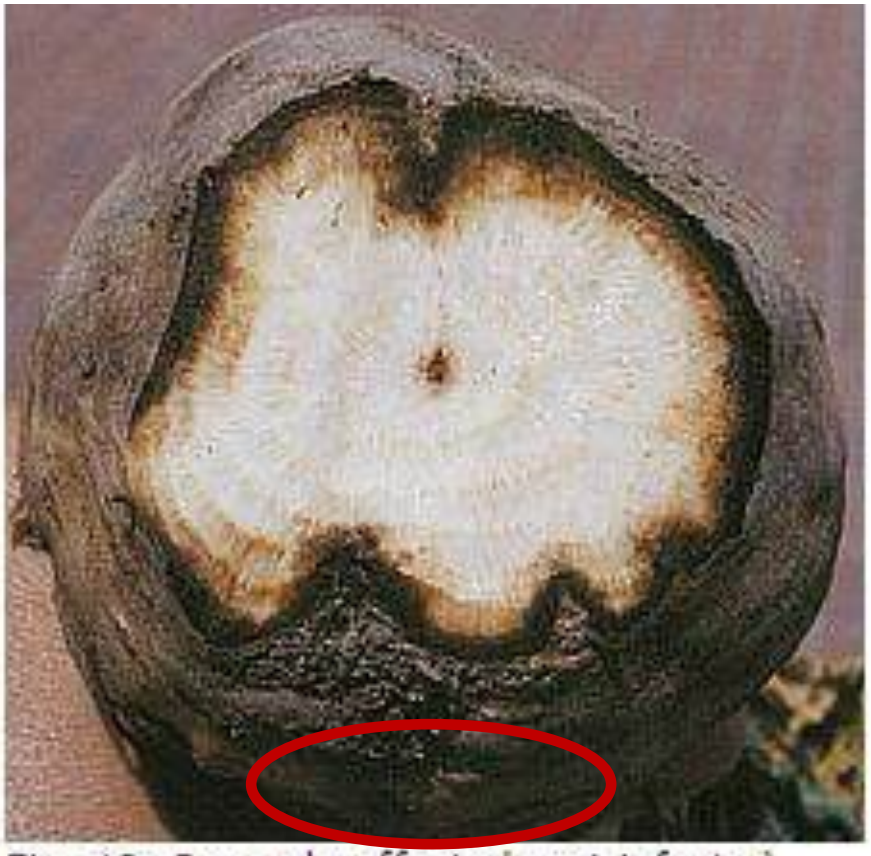


Fig. 19. Severely affected root infected with *R. solani*, showing rotting of external cortical tissues, with limited internal penetration.

Rhizoctonia root rot and blight – *Rhizoctonia solani*

Symptoms

General symptoms

- Poor germination and seedling emergence,
- Very uneven growth,
- Dead seedlings
- Reddish discoloration of above-ground plant parts .

Specific symptoms

- Seed decay and pre emergence damping-off,
- Post emergence damping-off
- Wire-stem
- Misshapen fleshy roots with external or internal rots.

Rhizoctonia root rot and blight – *Rhizoctonia solani*

Symptoms

On plants abnormal and Infected fleshy root

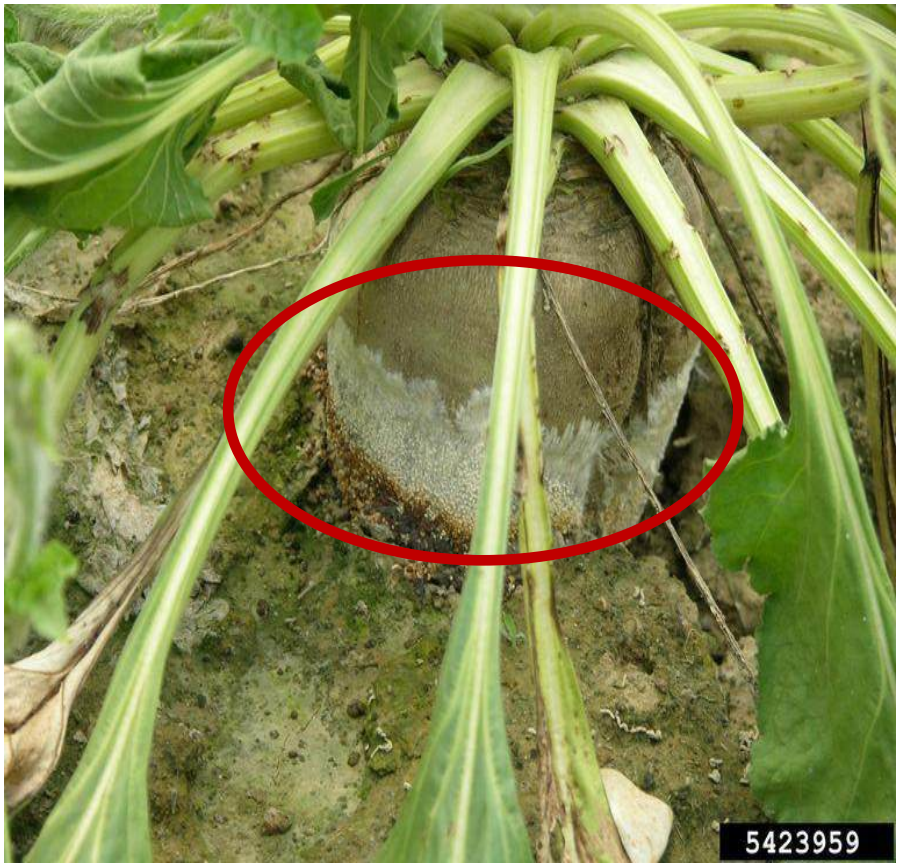
- Later in the growing season, infected plants that survive the **wire-stem phase develop abnormal fleshy roots** .
- Infected tissues of the root and stem enlarge more slowly than the surrounding healthy tissues, leading to **the formation of constrictions of various shapes** and sizes.
- At harvest, **infected fleshy roots may exhibit several dry rot symptoms**.
- The **rotted tissues are generally firm and dry, brown to black and sharply delimited by healthy tissue** .
- The rotted areas range in size from **a small lesion to the whole root.**
- Infection of the fleshy roots may also occur through the **petioles in the crown area** resulting in a downwardly progressing rot

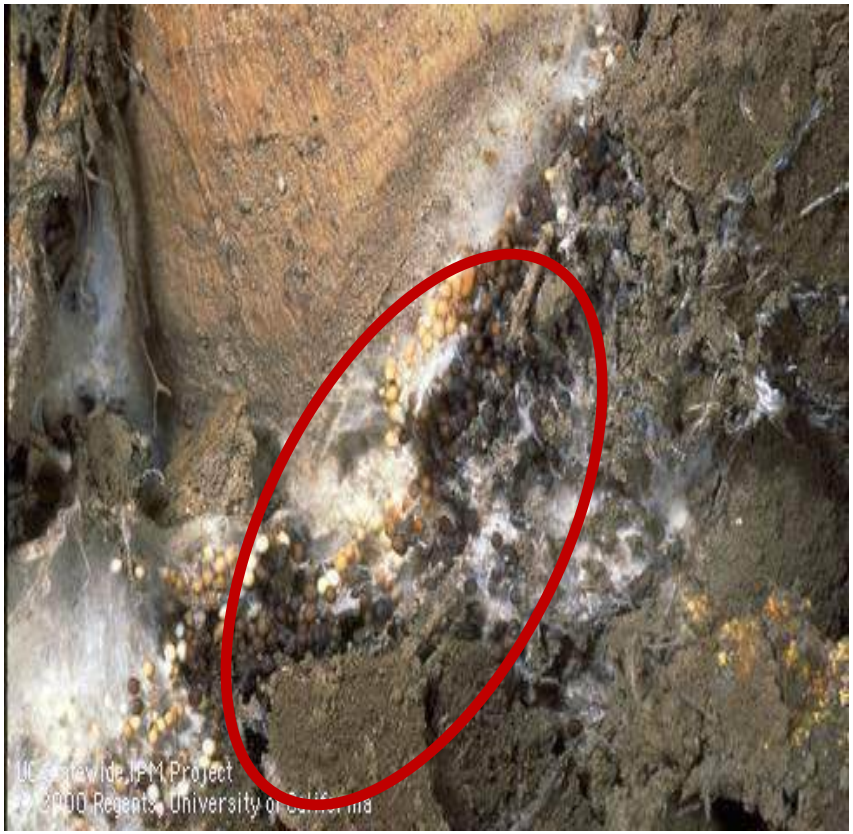
Rhizoctonia root rot and blight – *Rhizoctonia solani*

Management

- **Deep ploughing**
- **Crop rotation with non host crops like** corn, barley, wheat and oat (host crops - beans, cabbage, peas or potatoes)
- **Green manuring**
- Sowing on ridges will reduce damage by *Pythium*.
- Seed treatment with chemical fungicides (eg. carboxin,, iprodione, tolclofos-methyl, cyproconazole)
- Biocontrol - *Pseudomonas fluorescens*, *Trichoderma harzianum* for the control of *R. solani*

Sclerotium root rot of beet root - *Sclerotium rolfsii*





Sclerotium root rot of beet root - *Sclerotium rolfsii*

Symptoms

- The most important sign of the disease is the presence of white strands of fungal mycelium
- Numerous mustard seed sized sclerotia are found on the decaying parts
- **Yellowing and wilting of plants**
- The **fleshy roots are completely or partially decayed**
- Plants can be pulled out easily

Sclerotium root rot of beet root - *Sclerotium rolfsii*

Management

- Soil drenching with 0.3% copper oxychloride or 0.25% captan near the collar region.

- Drenching of 0.1% carbendazim should be initiated just prior to flowering stage or Thiophanate methyl (0.1%) or Combination of Mancozeb (0.25%) and Carbendazim.

- The fungicides hexaconazole, propiconazole, difenoconazole; combination products, avatar (hexaconazole 4% + zineb 68%), nativo (tebuconazole 50%+ trifloxystrobin 25%) and vitavax powder (thiram 37.5%+carboxin 37.5%)

Phoma blight and heart rot: ***Phoma betae***

Fig. 1. Symptoms of Phoma leaf spot (left) and Phoma root rot (right).

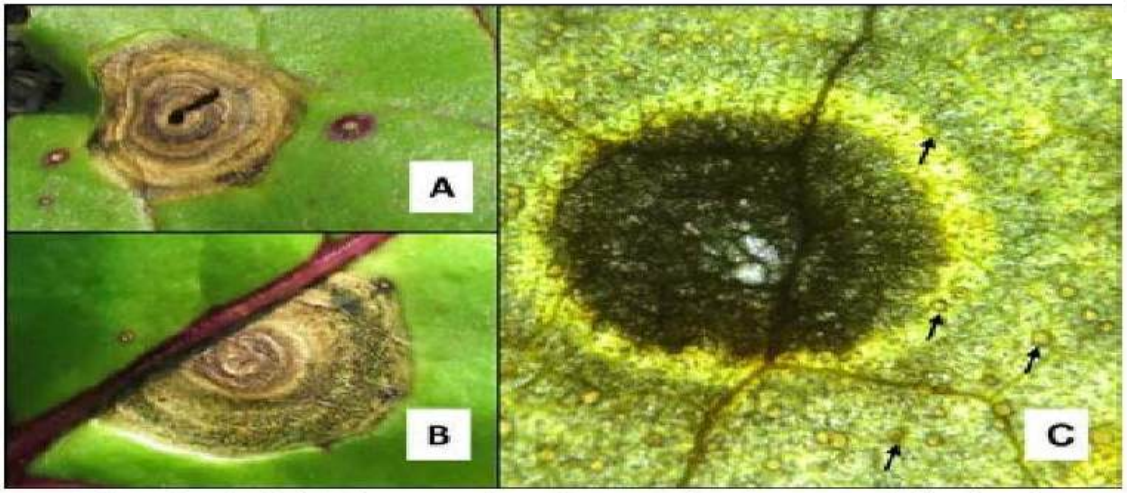


Fig. 2. A/B) Phoma leaf spot lesions with dark concentric rings; C) Close-up of lesion at 63X magnification with pycnidia (arrows).

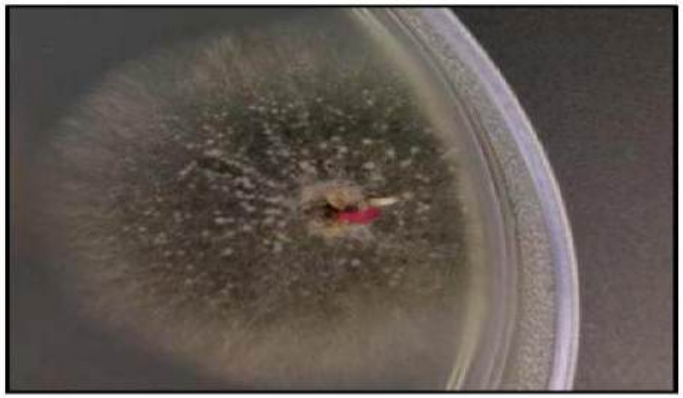


Fig. 3. Table beet seed infested with *Phoma betae*.

Phoma blight and heart rot: ***Phoma betae***

Phoma blight and heart rot: *Phoma betae*

Sugar beet

In seedlings - Rotting of fleshy roots

In mature plants - fungus become active results in dry black rot in sugar beet

Phoma blight and heart rot: ***Phoma betae***

- **Hosts:** Table beet, sugar beet, Swiss chard, spinach, lambsquarters (weed)

Symptoms:

- **More severe when there is boron or sodium is deficiency**
- Seed rot and damping off in young seedlings.
- **If survived the seedling stage infection, rotting of fleshy roots is observed.**
- Foliar symptoms include **tan-brown leaf spots with dark concentric rings .**
- Pycnidia produced in dark concentric rings within the lesion
- Root symptoms include **dark brown and dry lesions associated with shrinking.**
- Phoma root rot typically begins in the **center of the crown and spreads downward into the taproot (unlike Aphanomyces root rot).**

Phoma blight and heart rot: *Phoma betae*

Disease Cycle:

- *P. betae* is commonly introduced into fields through seed .

- The pathogen can be spread within a field through water-splashed conidia, surviving in the soil on infested plant debris, and on alternative weed hosts.

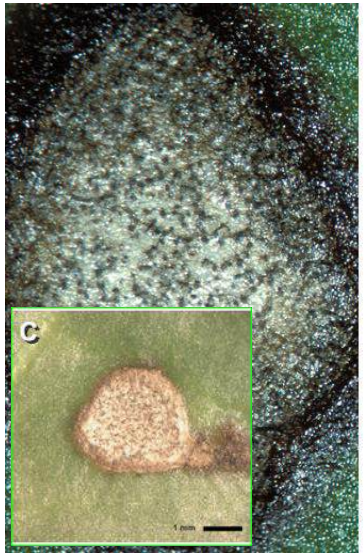
- The pathogen can be spread farther distances between fields through windblown ascospores.

Phoma blight and heart rot: *Phoma betae*

Management

- Use treated and certified seed
- Practice crop rotation (3 years)
- Dispose of plant residues after harvest
- Grow beets in well fertilized soils using clean seed
- **Hot water treatment of seeds at 59^0C for 8 min** followed by drying for 24 h and retreat with thiram @ 5g/kg seed
- Application of boron and sodium

Cercospora leaf spot (CLS) :*Cercospora beticola*



Cercospora leaf spot (CLS) :*Cercospora beticola*

DISEASE SYMPTOMS ARE INITIALLY DISCRETE, NECROTIC LESIONS SURROUNDED BY A RED TO PURPLE MARGIN IN A RED TABLE BEET CULTIVAR (FIG. 1). IN A YELLOW CULTIVAR, MARGINS OF LESIONS ARE TAN BROWN (FIG. 2). CLS LESIONS HAVE A GRAY CENTER ON ALL CULTIVARS.

Cercospora leaf spot (CLS) :*Cercospora beticola*

- This is most important disease affecting **table beet.**
- In processing table beet crops, epidemics cause **significant defoliation and may disrupt mechanized harvest.**
- For fresh market table beet production, **roots may be sold with or without intact foliage.**
- **The lesions on the foliage** may lead to rejection for fresh market sale.

Cercospora leaf spot (CLS) :*Cercospora beticola*

Symptoms

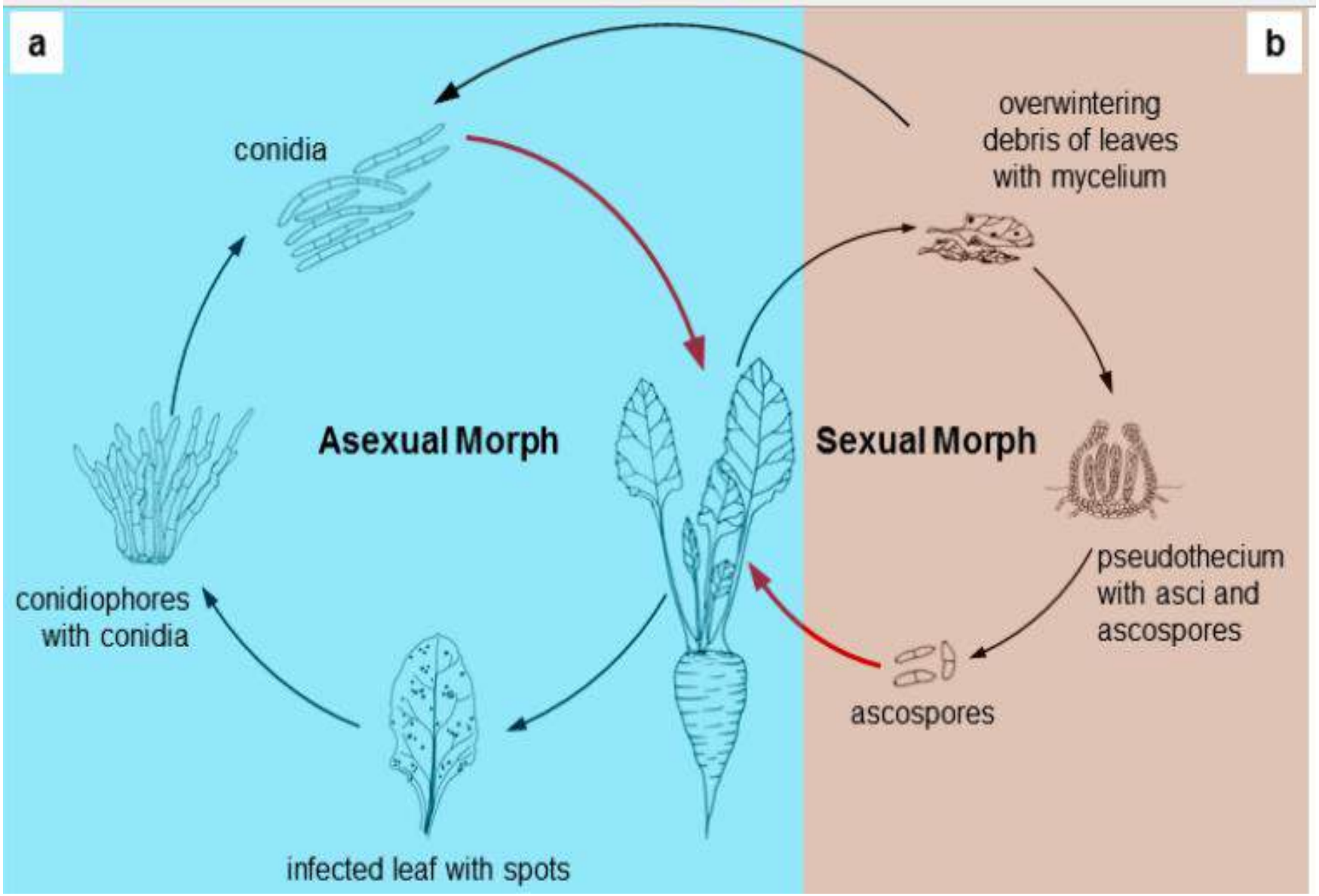
- The **lower leaves** near the ground are attacked first.
- Disease symptoms are initially discrete, necrotic lesions in red table beet cultivar lesions margins are **red to purple**
- In yellow cultivar **-** lesions margins are **tan brown .**
- Lesions have **a grey centre** on all cultivars
- Central tissue turns brown and assumes **grayish colour after sporulation**
- The spots dry up giving a shot hole appearance

Cercospora leaf spot (CLS) :*Cercospora beticola*

Symptoms contd…

- **Elongated spots** appear on petioles
- Spots coalesce, the leaves wrinkle and dry.
- As leaves die, the **crown becomes cone-shaped with a rosette of dead leaves at the base.**
- Defoliation occurs throughout the growing season resulting in **reduction in root size and yield.**
- **Older leaves are mostly affected**
- Reduction in root growth, yield and sugar content

Cercospora leaf spot - *Cercospora beticola*



Cercospora leaf spot - *Cercospora beticola*

Disease cycle

Alternate Hosts - swiss chard, spinach, and sugar beet.

Commonly encountered weeds Lambsquarters, Pigs weed also serve to perpetuate the fungus.

Soil and Plant Residues: *Cercospora beticola* is able to survive in the soil on infested plant residues for up to 3 years.

Cercospora leaf spot - *Cercospora beticola*

Management

- Reducing the periods of leaf wetness by decreasing planting densities and orienting rows parallel to the prevailing wind direction can be very helpful in minimizing the pathogen infection.
- Sanitization by removing infected leaves, plant debris and weeding can help manage disease.
- Mulching can be a good approach for limiting the weeds in the field
- Tillage methods also have been found to affect the disease development
- Spraying Chlorothalonil 1.5 ml/lit, Copper hydoxide 2g/l, Propiconazole 1.0 ml/lit ,Iprodione + Carbendazim 2g/l, Mancozeb+carbendazim 2g/l, Azoxystrobin 23% SC @ 0.1%, Kresoxim methyl 44.3% SC @ 0.1%, Pyraclostrobin + Metiram 60% WG @ 0.2%

Beet yellows – *Beet Yellows Virus, Beet Mild Yellowing Virus* (BMYV)

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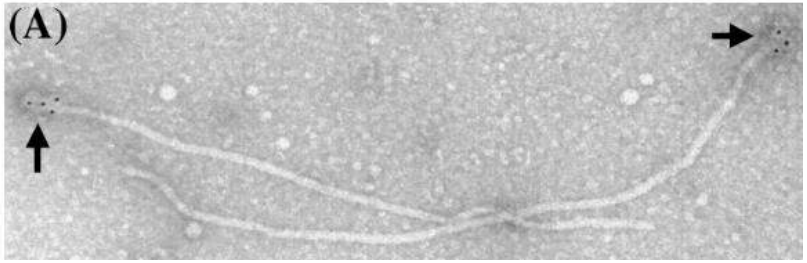
Beet yellows – *Beet Yellows Virus, Beet Mild Yellowing Virus* (BMYV)

Symptoms

- Mottling of leaves with occasional ring like lesions
- On garden beets, vein clearing
- Infected foliage is leathery, stunted and distorted
- Leaves can become thicker and then brittle
- Reduction in size of petioles
- Roots are smaller with hairy side roots

Beet yellows – *Beet Yellows Virus, Beet Mild Yellowing Virus* (BMYV



- **Gram +ve, ss RNA**
- **Closterovirus group.**
- ***Closteroviridae***
- **Longest viruses, Flexious rods**
- ***Beet Mild Yellowing Virus* make the plants more susceptible to fungal attack (Powdery mildew).**

Beet yellows – *Beet Yellows Virus, Beet Mild Yellowing Virus* (BMYV)

Survival and spread

P.I: Infected seed plants and weed hosts

S.I: Aphids (*Myzus persicae*, *Aphis fabae*, *Rhopalosiphum padi*

Beet yellows – *Beet Yellows Virus, Beet Mild Yellowing Virus* (BMYV)

Management

- Removal and destruction of weed hosts
- Plants selected for seed plants should be free of virus and grown in isolation
- Vector control with insecticides